



# NCCER Module 22107



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# Vertical-Mast Sit-Down Counterbalance Forklifts





# **Section Six**

## *Forklift Operation*

# Objective

6. Explain how to safely operate and use a forklift.
  - a. Explain the importance of preparing to work with a forklift.
  - b. Explain load capacity and its relationship to the Stability Triangle and Pyramid.
  - c. Describe how to start and shutdown a forklift.
  - d. Explain how to maneuver with a load.
  - e. Explain how to pick up, move, and place a load.



# Performance Tasks

Under the supervision of an instructor, trainees should be able to do the following:

5. Start, warm up, and shut down a forklift
6. Perform basic maneuvers with a forklift.
7. Perform basic lifting and placing operations with a forklift.



## 6.0.0 Forklift Operation

- Operating a forklift requires the worker to pay close attention to the controls and instruments, the load, and the surrounding environment.
- Operators must plan their work and movements in advance and be alert to other activities around their lift, especially if personnel are nearby.



## 6.1.0 Preparing to Work

- Board the lift by using the grab rails and foot rests while maintaining three points of contact.
- Once seated in the cab with your back against the seat back, adjust the seat to a comfortable operating position.
- Make sure you can easily reach and fully depress both the throttle and brake pedals. Make sure each of the controls can be reached easily.
- The operator should be fully familiar with the equipment, regardless of variations in the forklift brand or model.
- For IC-powered forklifts, a slow warm up is essential for proper operation of the machine, especially if it will be moving heavy loads.



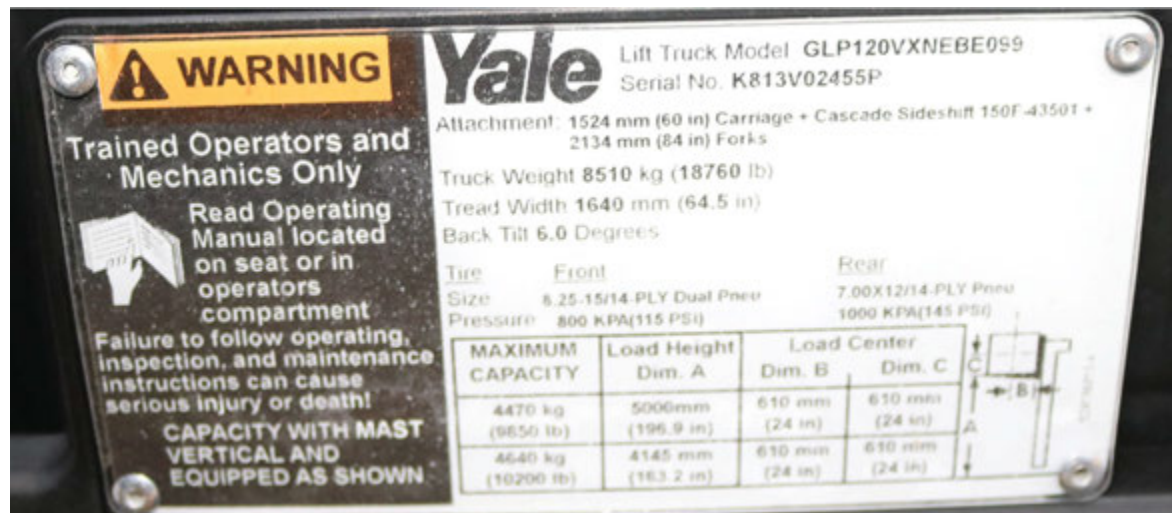
## 6.2.0 Load Basics

- Handling loads safely with a forklift is dependent on three factors: capacity, balance, and stability.
- Before operating the forklift, the operator should understand how these factors – individually and combined – impact a forklift's load.



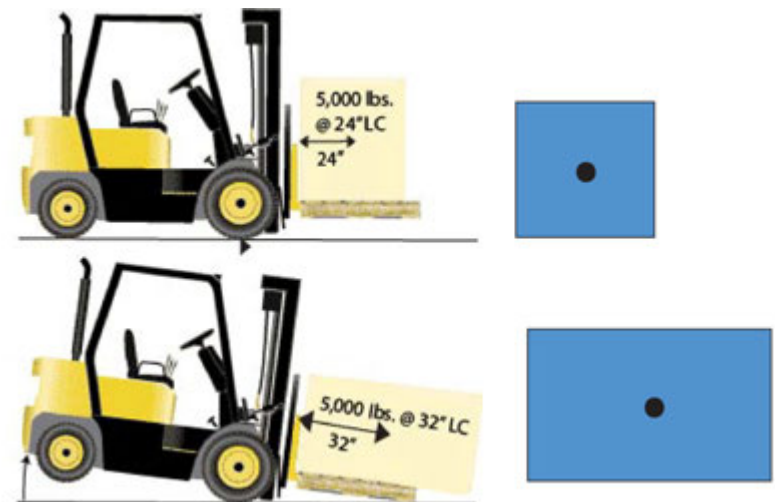
## 6.2.1 Capacity and Balance (1 of 2)

- The maximum load that a forklift can carry without tipping is the lift's rated capacity.
- Exceeding this weight jeopardizes the operator's safety, the safety of everyone near the forklift, the equipment, and the load.



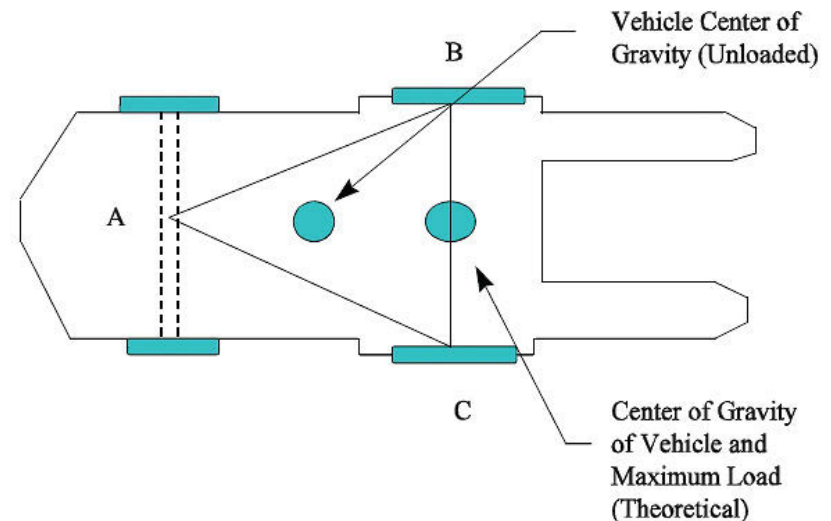
## 6.2.1 Capacity and Balance (2 of 2)

- A forklift will tip over if its rear counterbalance is not heavy enough to counteract the weight of the load.
- The further the load is away from the fulcrum, the harder it is to lift.
- The load capacity of a conventional forklift is the maximum weight the machine can lift based on a load's center.



## 6.2.2 Forklift Stability (1 of 3)

- A conventional forklift's stability is based on its three-points of support: each of the two front drive wheels and the center point of the steering axle.
- Connecting the three points with lines creates the **stability triangle**.
- A load is safe to carry when the forklift's center of gravity is within the stability triangle.



## 6.2.2 Forklift Stability (2 of 3)

- As a forklift raises a load, the combined center of gravity moves according to the load's weight and height and the forklift's altitude on each axis.
- When this occurs, the stability triangle is better represented as a **stability pyramid**.
- The forklift will tip over if the center of gravity touches or goes outside of the pyramid's edges.



## 6.2.2 Forklift Stability (3 of 3)

The path the center of gravity moves is directly tied to one or more of the three dimensions of forklift stability:

- ***Longitudinal Stability*** – Stability related to the center of gravity's movement from front to rear on the forklift. When a load is added on the forks, the center of gravity moves forward.
- ***Latitudinal Stability*** – Stability related to the center of gravity's lateral movement, or from side to side.
- ***Vertical Stability*** – Stability related to the height of the load, or how it moves up and down. As a load is lifted, it moves the center of gravity up and forward.



## 6.3.0 Basic Operations

Before starting the job, the operator should understand:

- Load size
- Forklift capacity
- The best route between pick-up and drop-off locations
- Obstacles that may interfere with vertical and horizontal clearances
- Potential need for a spotter



## ***6.3.1 Starting an IC-Powered Forklift (1 of 2)***

1. Always maintain 3 points of contact to board the forklift.
2. Be sure the F-N-R lever is in Neutral.
3. Ensure the parking brake is set.
4. Turn the ignition switch to the Start position.
5. Warm up the engine for at least 5 minutes. In colder temperatures, warm up the machine for a longer period of time.
6. Check all the gauges and instruments to ensure they are working properly.
7. Manipulate the control levers to ensure they are operating properly, and to warm up the hydraulic oil.



## ***6.3.1 Starting an IC-Powered Forklift (2 of 2)***

8. Step on the service brake then release the parking brake.
9. Shift the transmission into gear.
10. Check the service brake by holding the brake and stepping on the throttle.
11. Check the steering.
12. Make a final visual check for leaks, unusual noises, or vibrations before starting operation.



## 6.3.2 Checking Gauges at Start-up (1 of 2)

### **IC-powered forklifts:**

- Keep engine speed low until the oil pressure registers on the gauge.
- If the oil pressure light does not turn off within 10 seconds of starting, stop the engine, determine the problem, and take corrective action.
- Check the other gauges and indicators to confirm that the engine is functioning properly, paying close attention to the water temperature indicator and amp meter.
- If there are any problems, shut down the machine and investigate or get a mechanic to look at the problem.



## 6.3.2 Checking Gauges at Start-up (2 of 2)

### **Electric-powered forklifts:**

- The primary concern should be whether the level indicated on the battery charge indicator is full.
- If the battery has been attached to a charger overnight and the indicator shows the battery is not fully charged, determine the reason.
- Make sure the connection between the battery and power cable is not loose.
- If the connector is attached securely, confirm the battery acid level.
- If the reason cannot be determined, contact the manager.



### 6.3.3 Shutdown

1. Find a dry, level spot to park the forklift.
2. Lower the forks or attachments to the ground.
3. Idle down if using an IC-powered forklift and allow it to idle for 5 minutes.
4. Ensure the forklift is in neutral and set the parking brake.
5. Turn the engine start switch to the Off position on IC-powered forklifts.
6. Relieve any hydraulic pressure by moving the forklift operating levers back-and-forth until the forks are all the way down.
7. Shut off the batteries. If LPG powered, also shut off the cylinder valve.



## 6.4.0 Maneuvering

- Three key hazards to be aware of when driving a conventional forklift forward or backward are **tail swing**, **load swing**, and **swept area**.
- Tail swing occurs when making a turn while moving. The resulting turn causes the rear (or tail) of the forklift to sweep across the area behind it, called the swept area.
- The resulting tail swing can cause severe damage to anything hit by the rear of the forklift.
- Load swing occurs when traveling while making a turn. Doing so causes the forks – and load they are carrying – to swing across the area in front of the lift.



## ***6.4.1 Moving Forward***

1. Before moving, raise the forks approximately 12 inches above the ground. This should be the travel position of the forks whenever the forklift is moved.
2. Move the F-N-R lever to forward.
3. Release the parking brake.
4. Confirm that other workers are not nearby.
5. Press the accelerator pedal to start the forklift moving forward.



## ***6.4.2 Moving Backward***

1. If moving forward, always come to a complete stop.
2. Move the F-N-R lever to reverse.
3. Look in your intended direction of travel and clear any area where you will be going.
4. Press the accelerator pedal to start the forklift moving in reverse direction.



## 6.4.3 Steering and Turning

- The forklift uses its rear wheels – rather than the front wheels – to make a turn. The forklift's pivot point is the front wheel.
- The **spinner knob** is attached to the steering wheel and allows the operator to steer the forklift easily and quickly with one hand while using the other to control the hydraulic levers.



## ***6.4.4 Parking a Forklift***

**Avoid parking in or obstructing walkways, doorways, safety equipment, or on inclines.**

The following steps are used to park attended and unattended forklifts:

1. Lower the load and forks to the ground.
2. Set all controls in neutral.
3. Set the parking brake. (Chock the wheels if parked on an incline.) If the forklift will be unattended, shut it down.



## 6.5.0 Operational Movement

While most of the steps associated with forklift movement are straightforward, there are hazards that operators need to recognize before moving loads.



## ***6.5.1 Picking Up a Load (1 of 2)***

Before picking up a load, remember to check the position of the forks to ensure they are centered on the mast and spread as wide as the load will permit.

Use the following procedure to pick up a load:

1. Travel to the area at a safe rate of speed and keep the forks low and tilted slightly back.
2. Before picking up the load, make sure it is stable. Secure the load if it is unstable.
3. Approach the load squarely and center it evenly on the forks, remembering to account for the balance point.
4. Drive up to the load with the forks tilted slightly forward. Whether the load is on a pallet or blocks, make sure that the forks will clear both of them.



## 6.5.1 Picking Up a Load (2 of 2)

5. Move forward slowly until the leading edge of the load rests against the back of both forks.
6. Raise the carriage slowly until the forks contact the underside of the load.
7. Tilt the mast rearward to cradle the load against the backrest.
8. Continue raising the carriage until the load safely clears the ground.



## ***6.5.2 Picking Up an Elevated Load***

- Picking up a load from an elevated position is more difficult than picking up a load off the ground.
- Operators should use a signal person to help them in these situations.
- It is critical for operators to understand where their forks are in relation to the load.
- Elevated loads can fall farther and cause more damage to merchandise and personnel.



## 6.5.3 *Traveling with a Load (1 of 2)*

- Always travel at a safe rate of speed.
- Never travel with a load raised above 12 feet.
- Keep the carriage tilted rearward.
- Stay alert and watch the load and conditions in front of the lift.
- Alert others to your presence by honking the horn.



## ***6.5.3 Traveling with a Load (2 of 2)***

- Avoid sudden stops and abrupt turns.
- Be aware of the sweep and swing of the tail and load when turning.
- If forced to drive on a steep slope, keep the load as low as possible.
- Avoid traveling parallel to a slope.
- When driving up or down a slope with a load, make sure the forks are pointed uphill.
- If forced to turn on an incline, make the turn wide and slow; this minimizes the risk of tipping over.
- Avoid driving over bumps and holes.



## ***6.5.4 Placing a Load***

- Approach the drop zone with the forks tilted slightly back.
- Align the rear (near) edge of the load and lower it until the back of the load stops moving down.
- Slowly tilt the load forward until the front of the load stops moving down. The load is now resting on its bottom.
- Confirm that the front and back of the forks are clear of the load bottom.
- Back away slowly until the forks have cleared the load.



## ***6.5.5 Placing an Elevated Load***

1. Drive the forklift as close as possible to the landing point with the load kept low.
2. Raise the load slowly and carefully while maintaining a slight rearward tilt to keep the load cradled against the carriage backrest.
3. Once the load has been lifted to the correct height, slowly move forward until the load is in position.
4. Lower the forks until the load is safely resting on the landing.
5. Slowly back up the forklift until the forks have cleared the load and landing.
6. Lower the forks until they are at a safe traveling height.



## ***6.5.6 Unloading a Flatbed Truck (1 of 2)***

1. Position the forklift at either side of the truck bed.
2. Manipulate the control levers in order to obtain the appropriate fork height and angle.
3. Drive the forks into the opening of the pallet or under the loose material.
4. Adjust the controls, as required, to lift the material slightly off the bed.
5. Tilt the forks back to keep the pallet or other material resting against the carriage backrest.
6. Back the forklift away from the truck.



## ***6.5.6 Unloading a Flatbed Truck (2 of 2)***

7. Lower the forks to the travel position and move to the landing point.
8. Position the forklift so that the material can be placed in the desired area.
9. Lower the load until the material is placed and pressure on the forks is relieved.
10. Back the forklift away from the pallet.
11. Repeat the cycle until the truck is unloaded.



# Wrap Up – Trade Terms (1 of 2)

***Load swing:*** The action that occurs when an operator is traveling on a forklift and makes a turn, which causes the forks and load to swing across an area in front of the lift.

***Rollover Protective Structure (ROPS):*** A protective covering constructed from welded steel and designed to protect the forklift's operator from being crushed in a tipover situation.

***Spinner knob:*** An accessory attached to the steering wheel that allows the operator to steer the forklift with one hand while using the other to control the hydraulic levers.

***Stability pyramid:*** Formed by drawing imaginary lines from each of the three points on the stability triangle to a point that connects the lines at the peak of the forklift's lift range. The stability of the machine is impacted by the location of the forklift's center of gravity within the pyramid.



## Wrap Up – Trade Terms (2 of 2)

**Stability triangle:** A triangle formed by connecting with lines the forklift's two front drive wheels and the center point of the steering axle.

**Swept area:** The areas behind and in front of the forklift that are affected when an operator is traveling and makes a turn. Doing so creates areas swept by the tail and load.

**Tail Swing:** The action that occurs when an operator is traveling and makes a turn, which causes the rear (or tail) of the machine to swing across a space behind the forklift.



## 6.0.0 Section Review Question 1

1. How many points of contact must be maintained when climbing into the operator's cab?
  - a. one
  - b. two
  - c. three
  - d. four



## 6.0.0 Section Review Question 1 Answer

1. How many points of contact must be maintained when climbing into the operator's cab?
- a. one
  - b. two
  - c. three**
  - d. four



## 6.0.0 Section Review Question 2

2. Handling loads safely with a forklift is dependent upon understanding which three factors:
- a. Capacity, balance, and stability
  - b. Stability, height, and weight
  - c. Counterbalance, load weight, pivot point
  - d. Longitudinal stability, latitudinal stability, and vertical stability



## 6.0.0 Section Review Question 2 Answer

2. Handling loads safely with a forklift is dependent upon understanding which three factors:

- a. Capacity, balance, and stability**
- b. Stability, height, and weight
- c. Counterbalance, load weight, pivot point
- d. Longitudinal stability, latitudinal stability, and vertical stability



## 6.0.0 Section Review Question 3

3. The stability triangle is created when imaginary lines are used to connect \_\_\_\_\_.
- a. the forklift's center of gravity with the top of the mast and the forklift's fulcrum
  - b. the counterbalance and both back wheels
  - c. each of the two front drive wheels and the center point of the steering axle
  - d. the forklift's fulcrum, top of the load backrest, and counterbalance



## 6.0.0 Section Review Question 3 Answer

3. The stability triangle is created when imaginary lines are used to connect \_\_\_\_\_.
- a. the forklift's center of gravity with the top of the mast and the forklift's fulcrum
  - b. the counterbalance and both back wheels
  - c. each of the two front drive wheels and the center point of the steering axle**
  - d. the forklift's fulcrum, top of the load backrest, and counterbalance



## 6.0.0 Section Review Question 4

4. At start up, if the oil pressure light does not turn off within 10 second of starting, the operator should\_\_\_\_\_.
- a. stop the engine, try restarting it, then resume operation if the light stays off
  - b. stop the engine, determine the problem, and take corrective action
  - c. set the brake then add oil while the engine is running until the light turns off
  - d. park the forklift and schedule preventive maintenance



## 6.0.0 Section Review Question 4 Answer

4. At start up, if the oil pressure light does not turn off within 10 second of starting, the operator should\_\_\_\_\_.
- a. stop the engine, try restarting it, then resume operation if the light stays off
  - b. stop the engine, determine the problem, and take corrective action**
  - c. set the brake then add oil while the engine is running until the light turns off
  - d. park the forklift and schedule preventive maintenance



## 6.0.0 Section Review Question 5

5. Two key hazards to be aware of when driving a conventional forklift forward or backward are \_\_\_\_\_.

- a. load weight and height
- b. mast height and fork length
- c. tail and load; sweep and swing
- d. running and brake lights



## 6.0.0 Section Review Question 5 Answer

5. Two key hazards to be aware of when driving a conventional forklift forward or backward are \_\_\_\_\_.

- a. load weight and height
- b. mast height and fork length
- c. tail and load; sweep and swing**
- d. running and brake lights



## 6.0.0 Section Review Question 6

6. Loads should not exceed the height of the \_\_\_\_\_.

- a. top of the lowered mast
- b. top of the carriage backrest
- c. lowest doorway
- d. top of the overhead guard



## 6.0.0 Section Review Question 6 Answer

6. Loads should not exceed the height of the

\_\_\_\_\_.

- a. top of the lowered mast
- b. top of the carriage backrest**
- c. lowest doorway
- d. top of the overhead guard



**Next...**



## **Module Examination and Review**

*Complete the Review Questions at the end of the module and prepare for the Module Exam.*